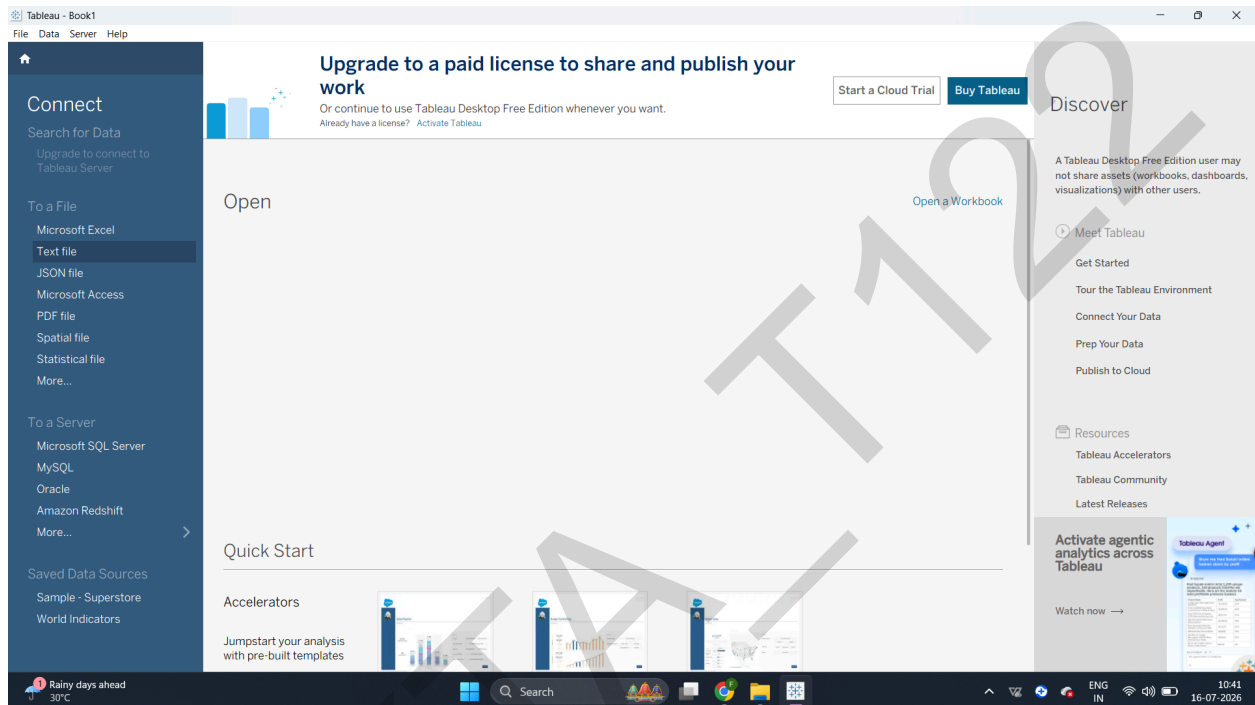


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Aim : Working with Data Sources and Data Preparation in Tableau

Step A: Connecting to Different Data Sources

1. Launch the Tableau Desktop application.
2. Locate the blue Connect control panel on the far-left section of the start page.



Step B: Creating Data Extracts vs. Live Connections

1. Once the dataset populates the main staging view, navigate to the top-right corner of the Data Source interface.
2. Locate the connection mode toggle section.

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Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders.csv

8 fields 99441 rows

100 rows

Name Orders.csv

Description

No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Order Status	Order...	order_...

Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approv
e481f51cbdc54678b7cc4913...	9ef432eb6251297304e7618...	delivered	02-10-2017 10:56:00	02-10-2017 11
53cdb2fc8bc7dce0b6741e21...	b0830fb4747a6c6d20dea0b...	delivered	24-07-2018 20:41:00	26-07-2018 0
47770eb9100c2d0c44946d...	41ce2a54c0b03bf3443c3d9...	delivered	08-08-2018 08:38:00	08-08-2018 C
949d5b44dbf5de918fe9c16f...	f88197465ea7920adcdbe73...	delivered	18-11-2017 19:28:00	18-11-2017 19:
ad21c59c0840e6cb83a9ceb...	8ab97904e6daea8866dbdbc...	delivered	13-02-2018 21:18:00	13-02-2018 2:
a4591c265e18cb1dcee52889...	503740e9ca751ccdda7ba28e...	delivered	09-07-2017 21:57:00	09-07-2017 2:

Go to Worksheet

Data Source Sheet 1

Rainy days ahead 30°C

Search

ENG IN

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3. Switch the active selection from Live over to Extract.

Tableau - Book1

File Data Server Window Help

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Order Id	Customer Id	Order Status	Order Purchase Timestamp	Order Approv
e481f51cbdc54678b7cc4913...	9ef432eb6251297304e7618...	delivered	02-10-2017 10:56:00	02-10-2017 11
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47770eb9100c2d0c44946d...	41ce2a54c0b03bf3443c3d9...	delivered	08-08-2018 08:38:00	08-08-2018 C
949d5b44dbf5de918fe9c16f...	f88197465ea7920adcdbe73...	delivered	18-11-2017 19:28:00	18-11-2017 19:
ad21c59c0840e6cb83a9ceb...	8ab97904e6daea8866dbdbc...	delivered	13-02-2018 21:18:00	13-02-2018 2:
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Go to Worksheet

Data Source Sheet 1

30°C Mostly cloudy

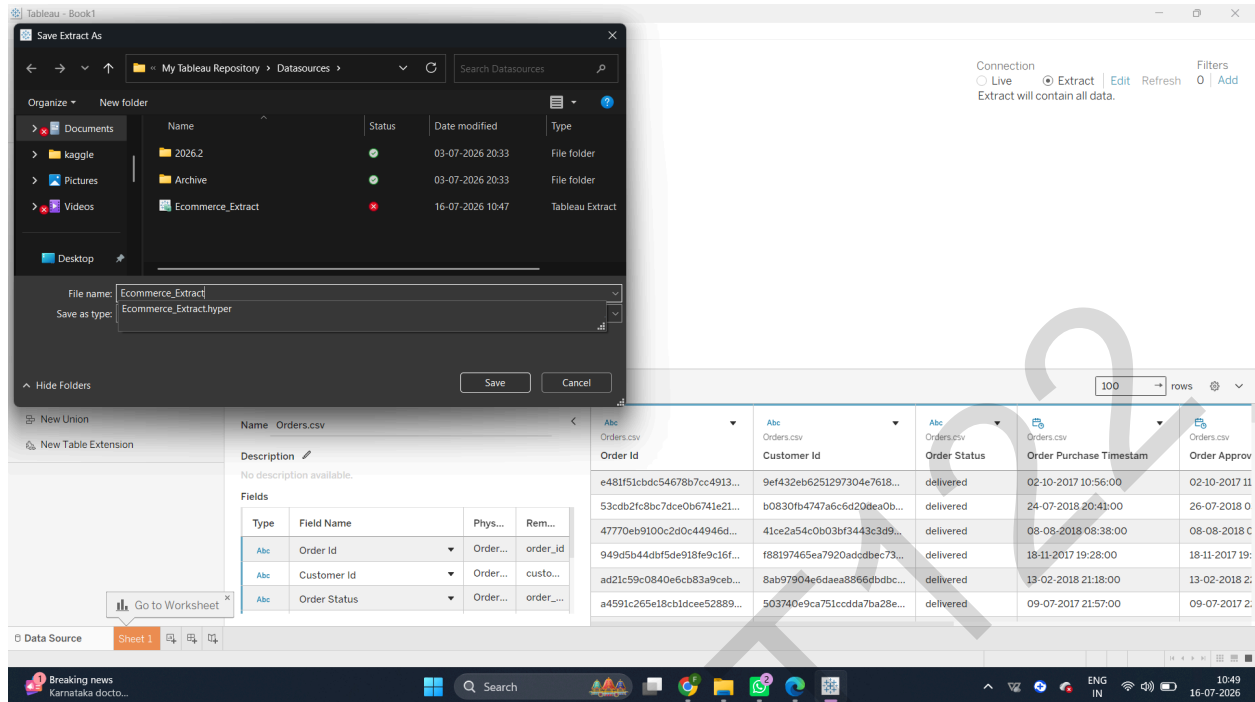
Search

ENG IN

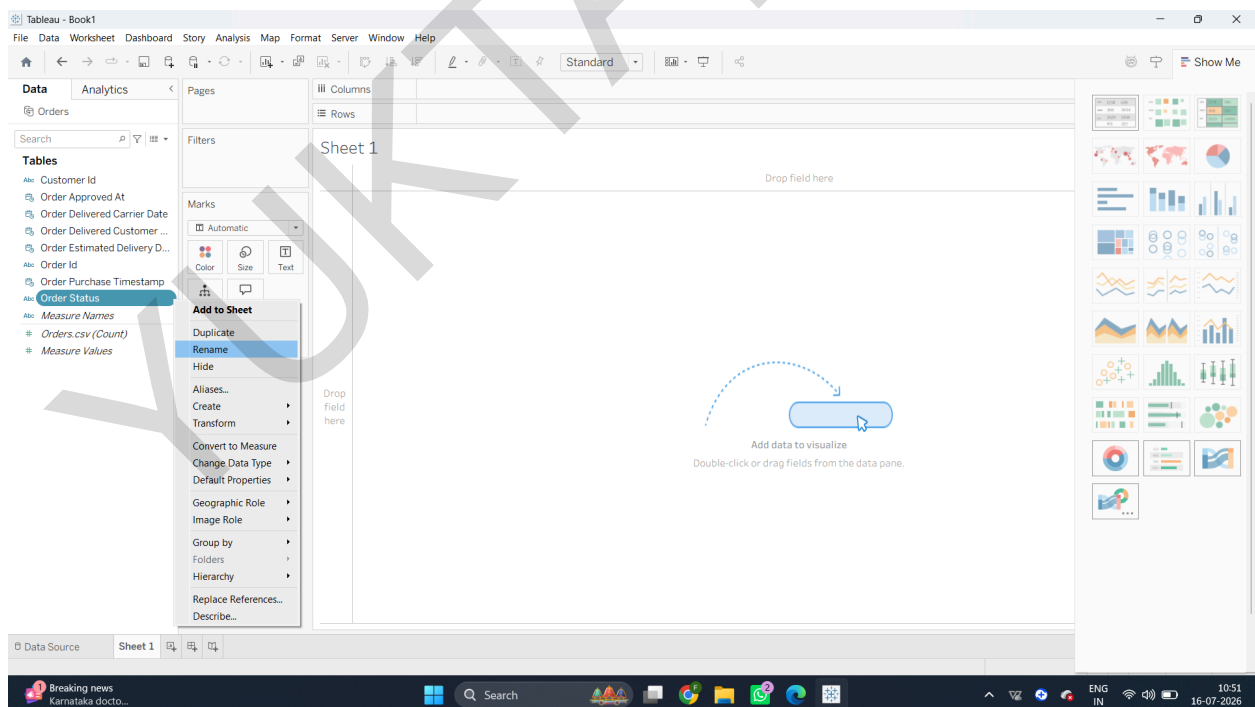
1046 16-07-2026

4. Click on the Sheet 1 navigation tab located at the bottom-left corner of the workspace screen. 5. A system dialog save window will immediately appear. Name the local snapshot file Ecommerce_Extract and click Save.

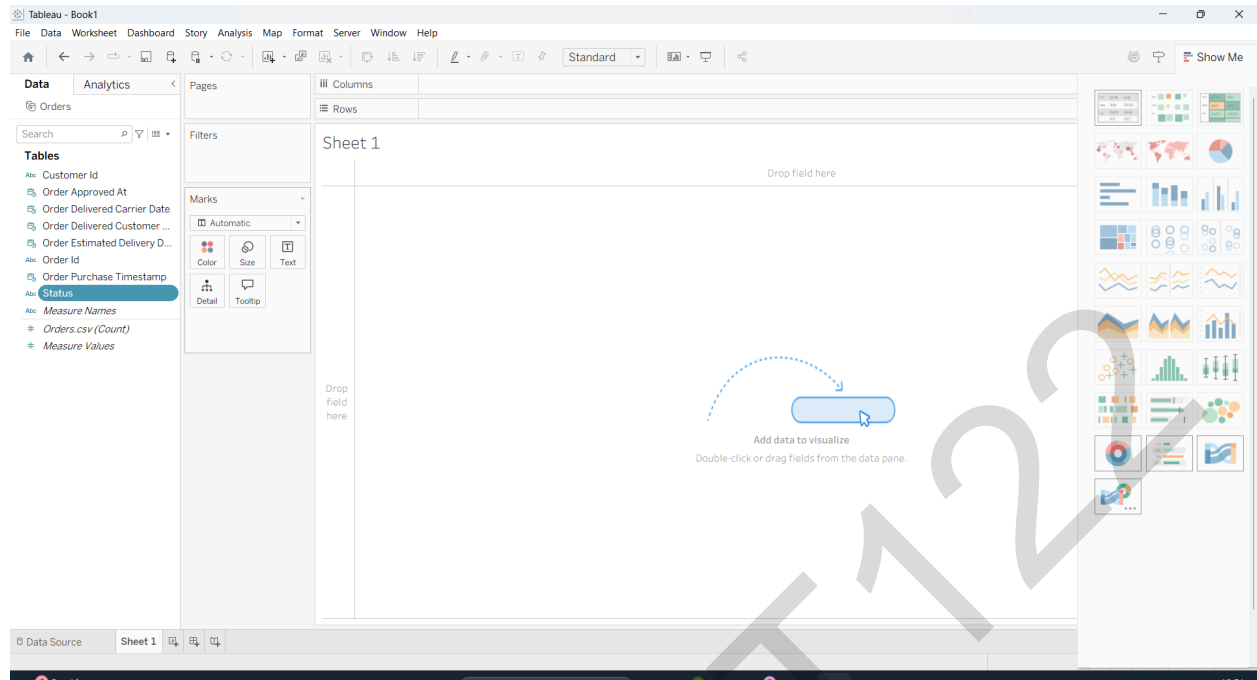
SHETH L.U.J. AND SIR M.V. COLLEGE



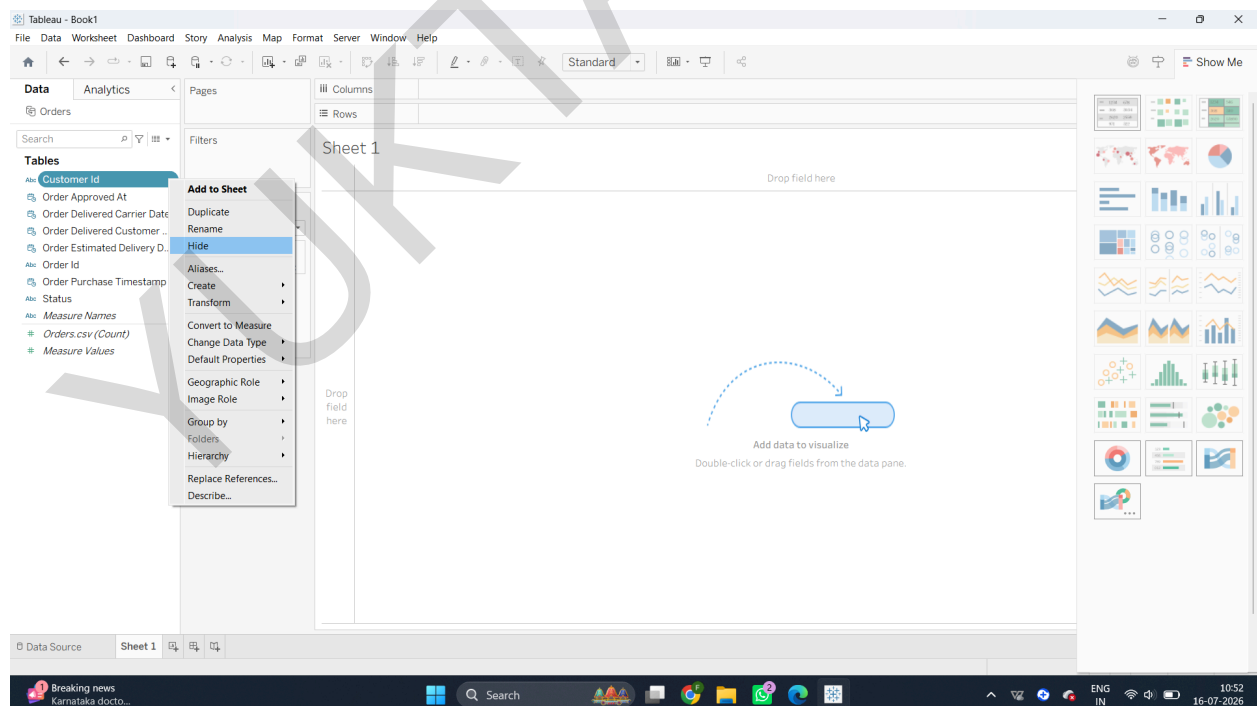
Step C: Managing Operational Metadata 1. Navigate to the blank workspace canvas on Sheet 1 and view the left-hand side Data Pane. 2. Rename a Field: Locate the dimensional attribute field labeled `order_status` or Order Status. Right-click the field name, select Rename, type in Status, and hit Enter.



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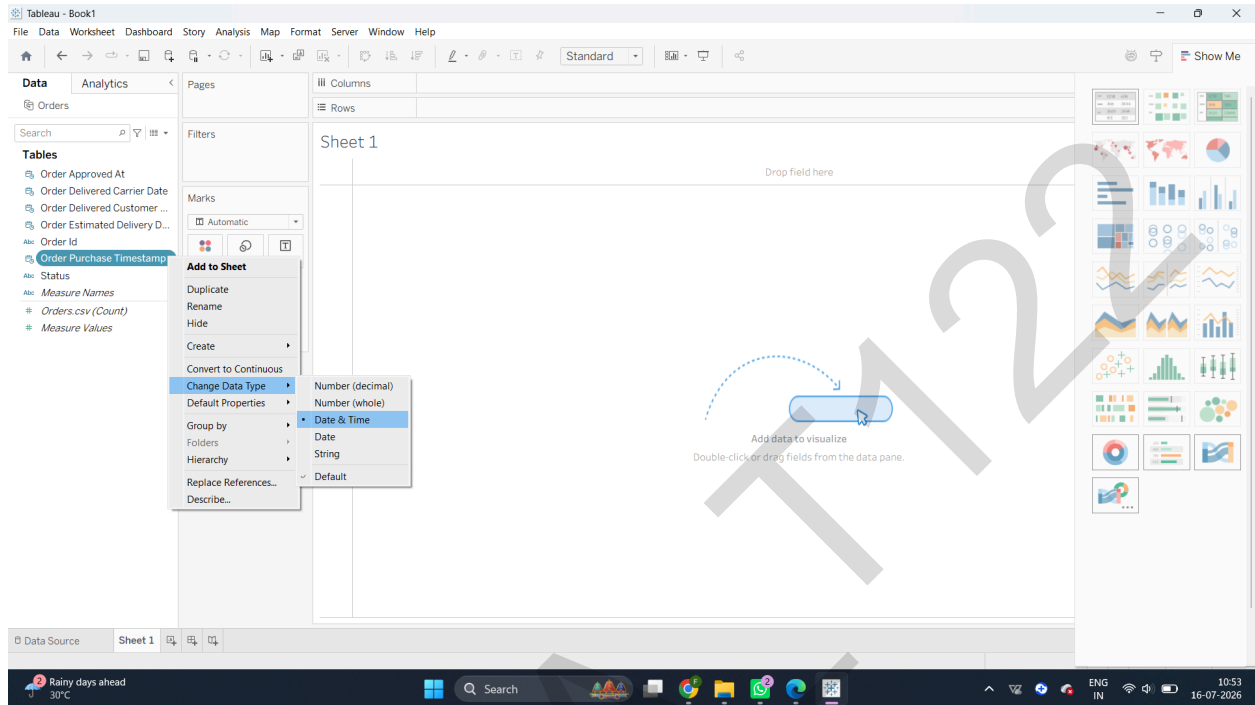


3. Hide a Field: Locate an internal tracking string variable that will not be used in visual reporting, such as customer. Right-click the field name and select Hide.

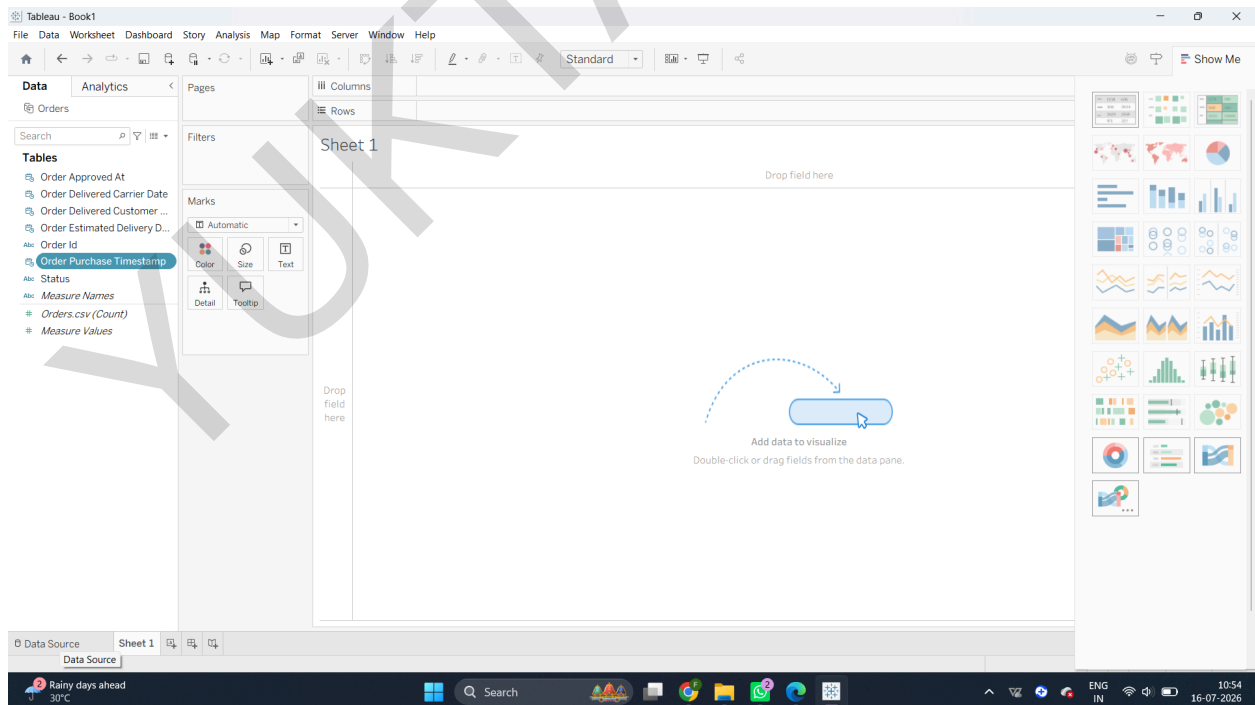


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4. Change a Data Type: Locate the operational timestamp column labeled `order_purchase_timestamp`. Click directly on the data type icon (which may incorrectly display as text `Abc` or a calendar date) positioned immediately to the left of the field name. Select `Date & Time` from the context dropdown menu to force proper structural recognition (Note: It may by default selected to `Date and Time`).



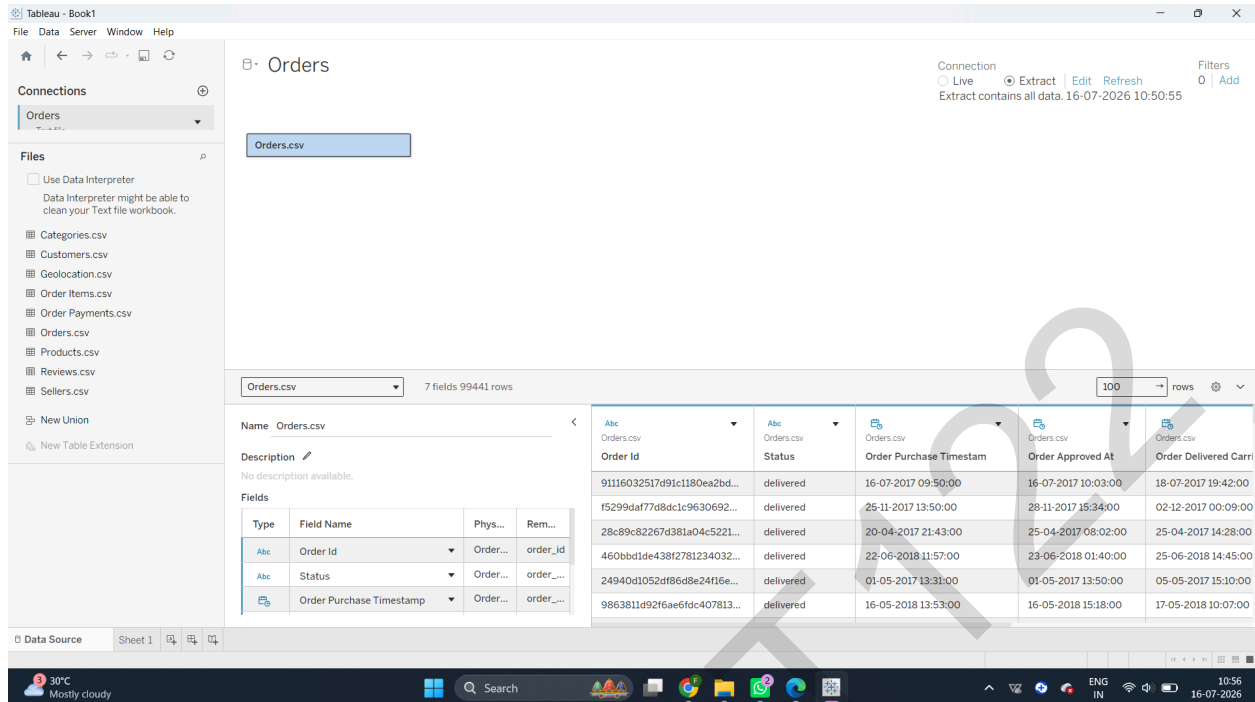
Step D: Performing Relational Joins on Multiple Tables 1. Click the `Data Source` tab positioned at the bottom-left corner of the window to return to the data schema designer.



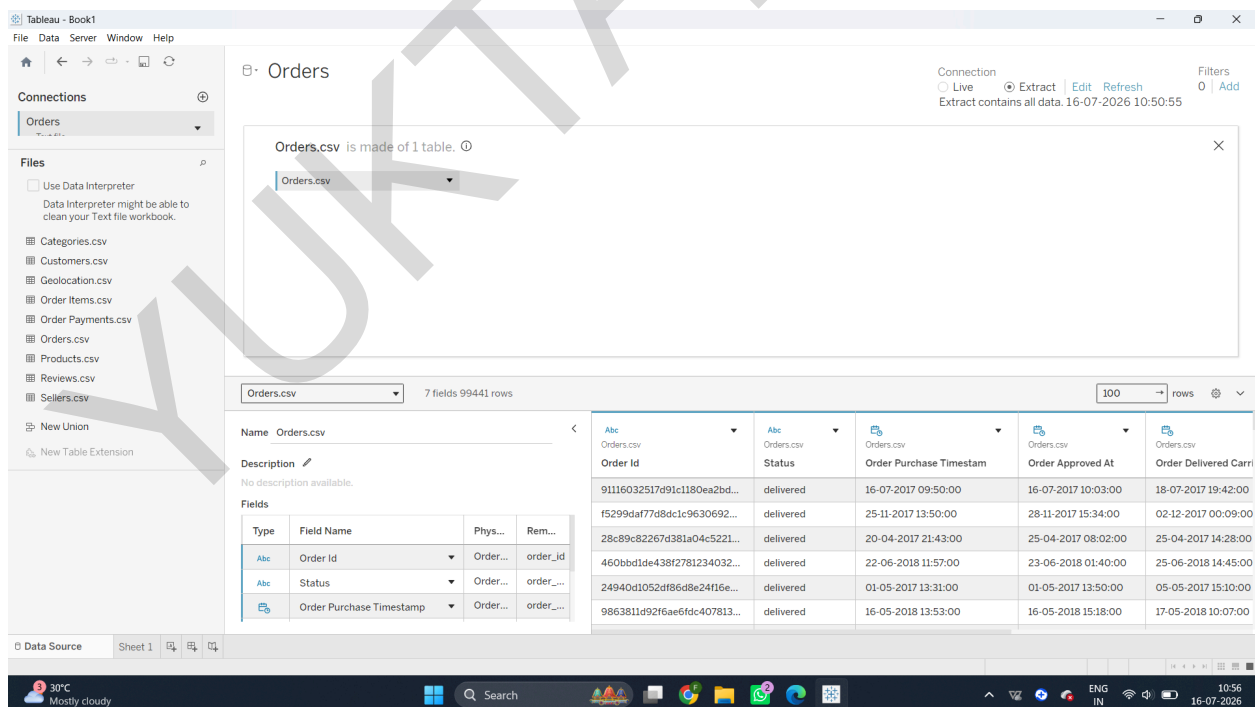
2. In the top staging canvas area, locate the rectangular `Orders.csv` database block.

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3. Double-click directly on the Orders.csv block. This action drills down through the logical relationship layer and enters the Physical Layer canvas space.



4. From the left-hand files listing pane, click and drag the Order Payments.csv file, dropping it directly next to the active Orders.csv block. A traditional Venn diagram icon link will snap open between them.

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Tableau - Book1

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New Union
New Table Extension

Orders

Connection: Live, Extract, Edit, Refresh, Filters: 0, Add

Orders.csv is made of 2 tables.

Orders.csv

Order Payments.csv

12 fields 103886 rows

100 rows

Name	Orders.csv		
Description	No description available.		
Fields			
Type	Field Name	Phys...	Rem...
Order Id	Order...	order_id	
Status	Order...	order_...	
Order Purchase Timestamp	Order...	order_...	

Order Id	Status	Order Purchase Timestamp	Order Approved At	Order Delivered Carri
b81ef226f3fe1789b1e8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aefd127...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93396b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcd1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42fd880ba16b47b5925idd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00

5. Click directly on the Venn diagram icon to open the Join configuration properties box.

Tableau - Book1

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42fd880ba16b47b5925idd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386b001f06208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00
3d7239c394a212fae122962...	delivered	05-06-2017 10:45:00	05-06-2017 10:55:00	14-06-2017 10:26:00
1f78449c87a54faf9e96e88b...	delivered	22-07-2018 14:07:00	23-07-2018 11:32:00	23-07-2018 12:58:00
0573b5e23cbd798006520e1...	delivered	04-07-2017 20:32:00	06-07-2017 03:15:00	06-07-2017 13:27:00

6. Under the data grid coupling properties, select order_id as the matching data key column drop-down for both the left and right tables (Note: in some case Tableau automatically selects the Order Id).

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7. Toggle Join Configurations: o Click Inner: Observe the underlying data preview window at the bottom. The rows instantly filter to show only records that contain both an order entry and a corresponding transaction payment receipt.

The screenshot shows the Tableau interface with the 'Orders' worksheet. The 'Connections' pane on the left lists various data sources. The 'Files' pane shows a list of CSV files. The 'Orders.csv' data source is selected, showing 12 fields and 103886 rows. The 'Join' dialog box is open, showing the 'Inner' join configuration between 'Orders.csv' and 'Order Payments.csv'. The join condition is 'Order Id = order id (Order ...)'. The data preview table shows the following columns: Order Id, Status, Order Purchase Timestamp, Order Approved At, and Order Delivered Carri. The table contains 10 rows of data, all of which have matching Order IDs in both tables.

Order Id	Status	Order Purchase Timestamp	Order Approved At	Order Delivered Carri
42fd880ba16b47b59251dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00
771ee386001f06208a7419e...	delivered	23-06-2017 13:11:00	23-06-2017 13:25:00	26-06-2017 17:33:00
3d7239c394a212faae122962...	delivered	05-06-2017 10:45:00	05-06-2017 10:55:00	14-06-2017 10:26:00
1f78449c87a54faf9e96e88b...	delivered	22-07-2018 14:07:00	23-07-2018 11:32:00	23-07-2018 12:58:00
0573b5e23cbd798006520e1...	delivered	04-07-2017 20:32:00	06-07-2017 03:15:00	06-07-2017 13:27:00

o Click Left: Notice the view updates to safely preserve 100% of the root rows coming from Orders.csv, leaving payment parameter cells completely empty if an order has no matching payment record.

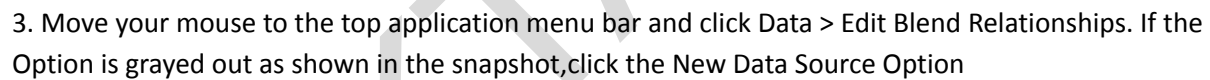
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Order Id	Status	Order Purchase Timestamp	Order Approved At	Order Delivered Carri
b81ef226f3fe1789b1e8b2aca...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00	02-05-2018 15:20:00
a9810da82917af2d9aef1d27...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00	28-06-2018 14:18:00
25e8ea4e93396b6fa0d3dd7...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00	15-12-2017 20:13:00
ba78997921bbcd1373bb41e...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00	07-12-2017 20:28:00
42fd880ba16b47b59251dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00	22-05-2018 11:46:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00	10-05-2018 13:35:00

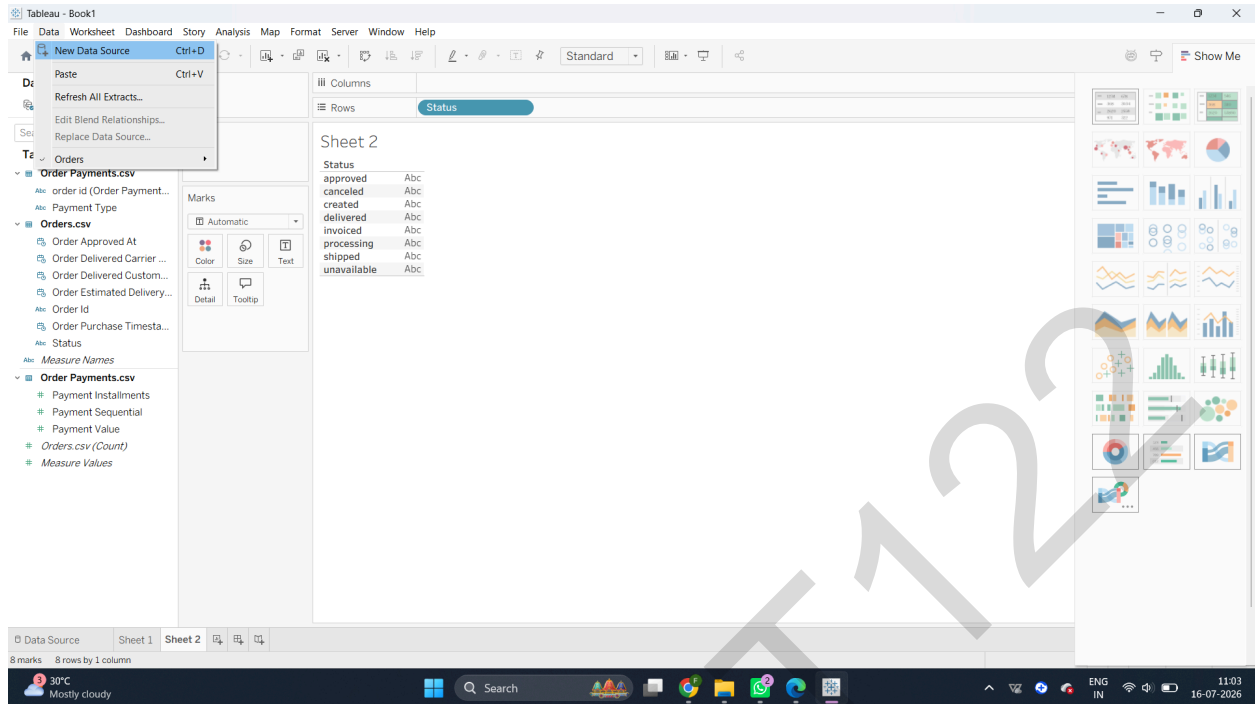
8. Leave the configuration set to a Left Join to preserve complete order history logs for the remainder of this exercise.

Step E: Applying Data Blending Between Independent Sources

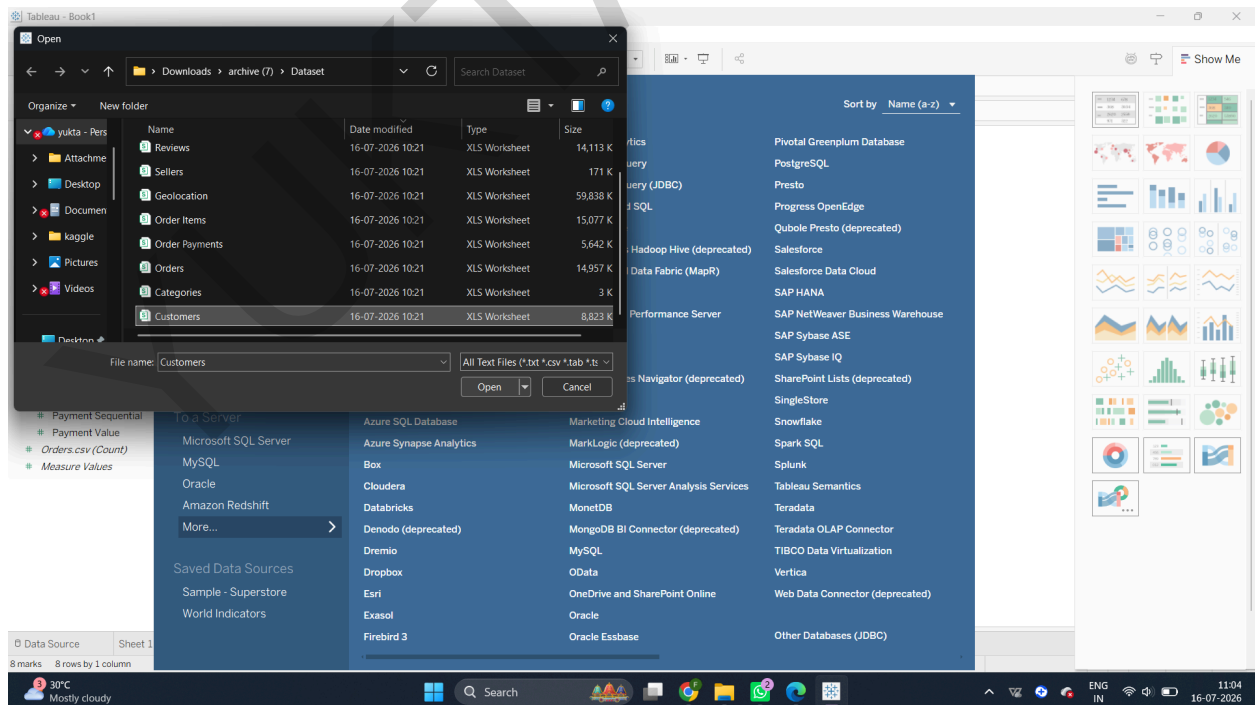
1. Create a brand new blank workspace sheet by clicking on the New Worksheet icon (the chart block containing a \$+\$ symbol) positioned at the bottom navigation tab area. This opens Sheet 2.
2. Ensure your joined Orders dataset connection is highlighted at the top left of the Data Pane. Drag the newly modified Status dimension field and drop it onto the Rows shelf. A blue checkmark will appear on the source, designating it as the Primary data pipeline.



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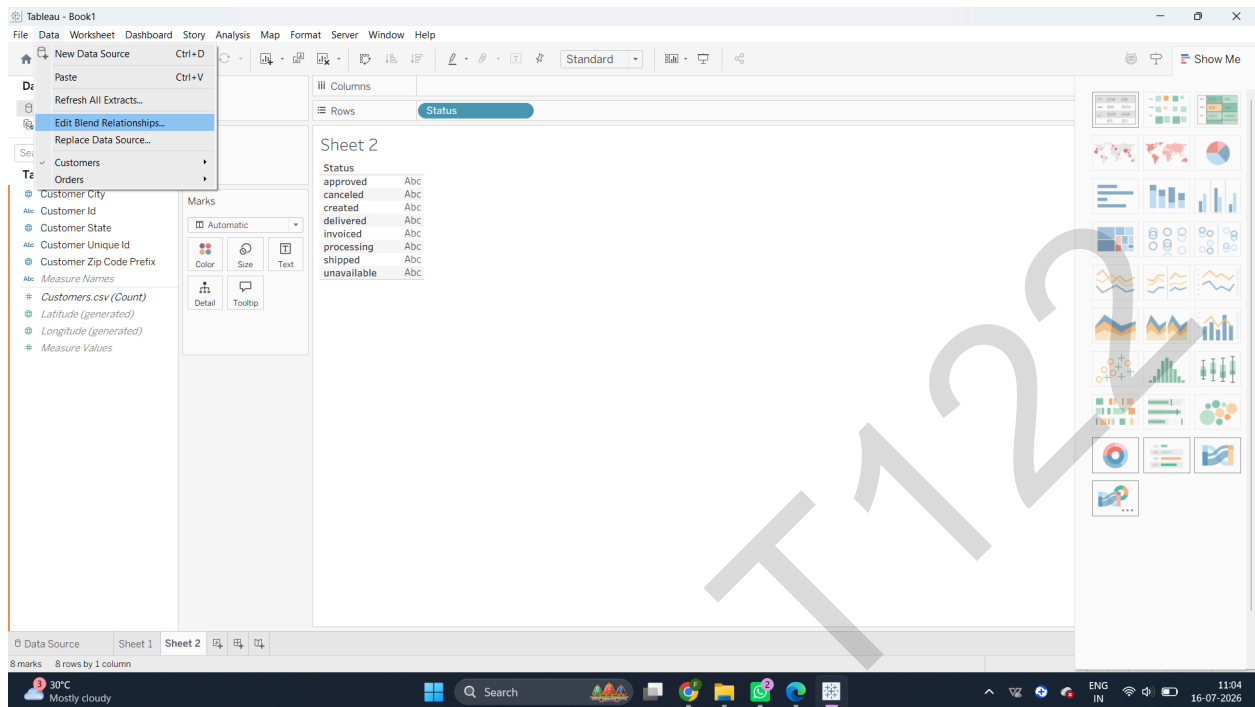


After Clicking new data source you are directed to blue menu window select text file to add the Customers.csv file.



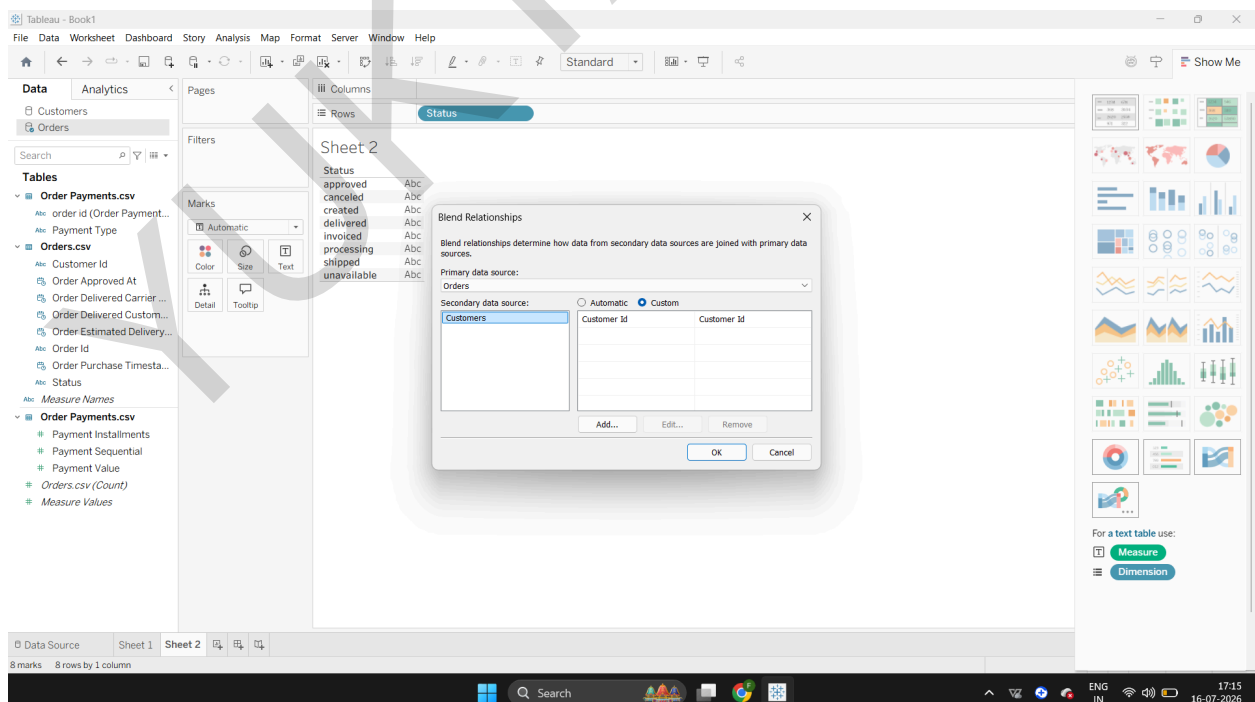
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After adding the CSV file get back to sheet 2 and recheck the Edit Blend Relationship Option and click on it



4. Set the Primary data source field box to your active Orders connection, and map the Secondary data source block to the independent Customers dataset file.

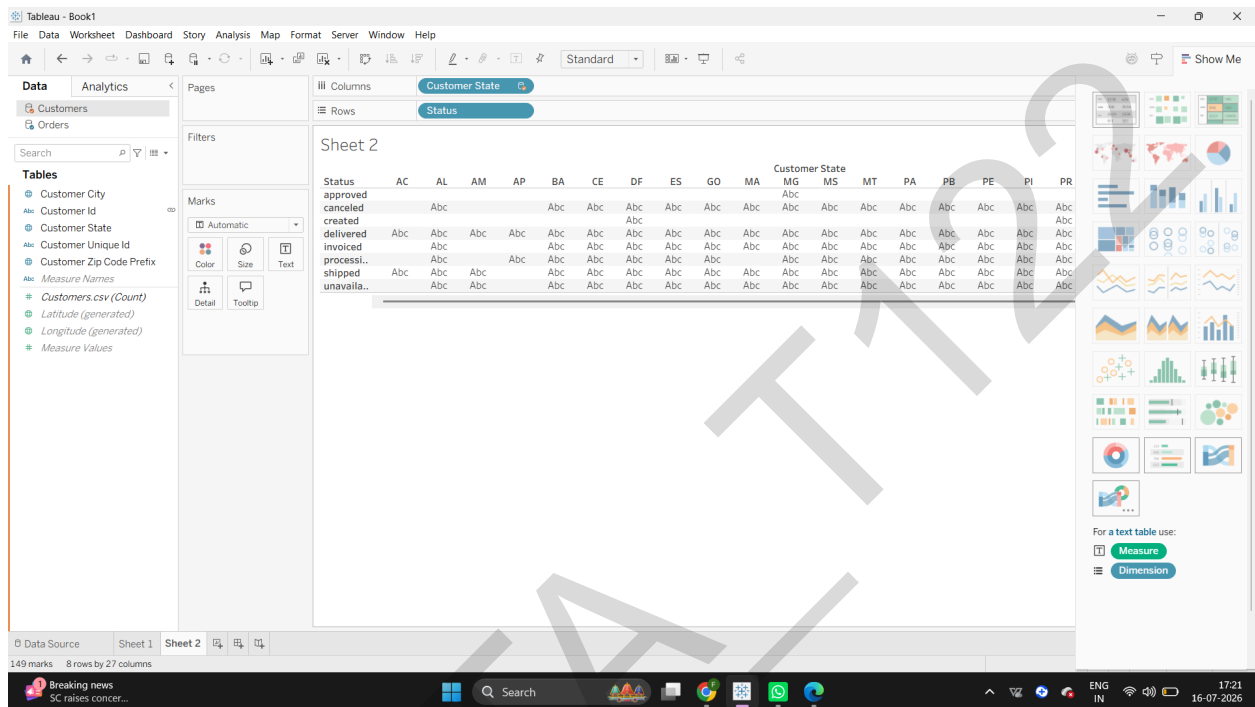
5. Switch the selection toggle switch from Automatic over to Custom, click the Add... button on the right, pair customer_id from the left dataset column to customer_id in the right column, and click OK.



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6. Now, go back to the top-left section of your Data Pane and click directly on the separate Customers dataset tracking label. An orange chain link icon will now appear locked together immediately next to the customer_id field.

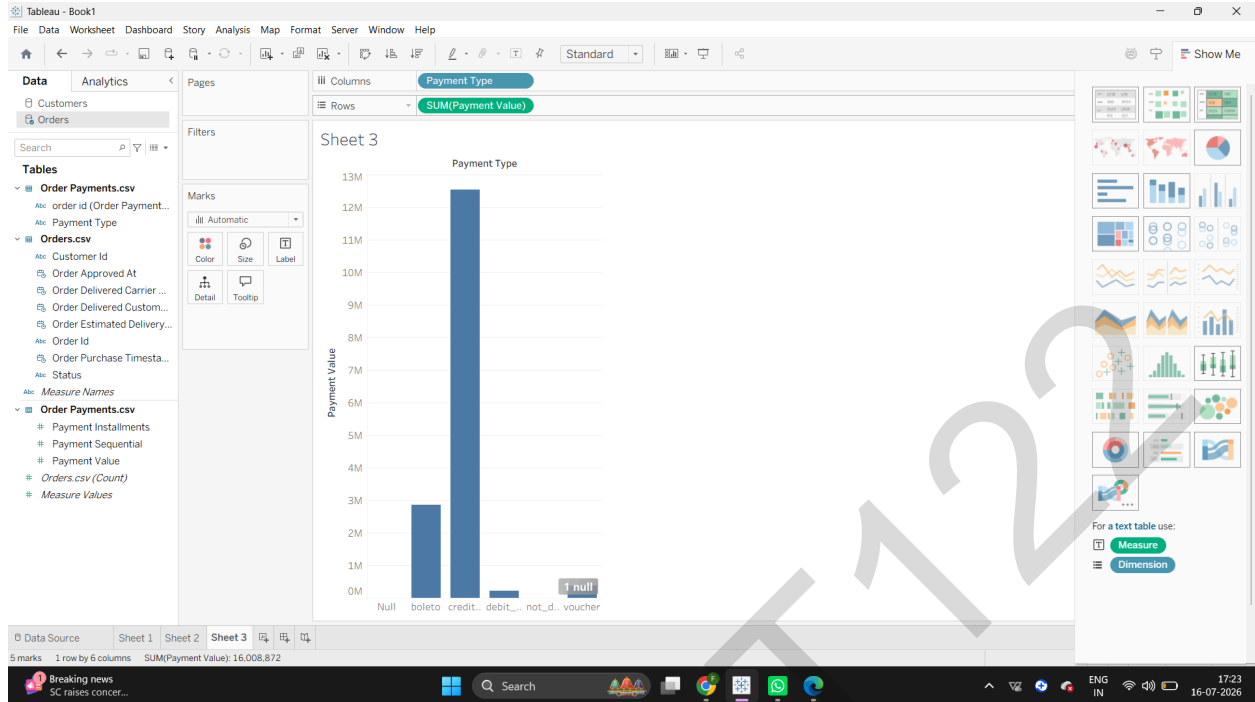
7. Click and drag the customer_state dimension variable out from this secondary dataset list and drop it onto the Columns shelf.



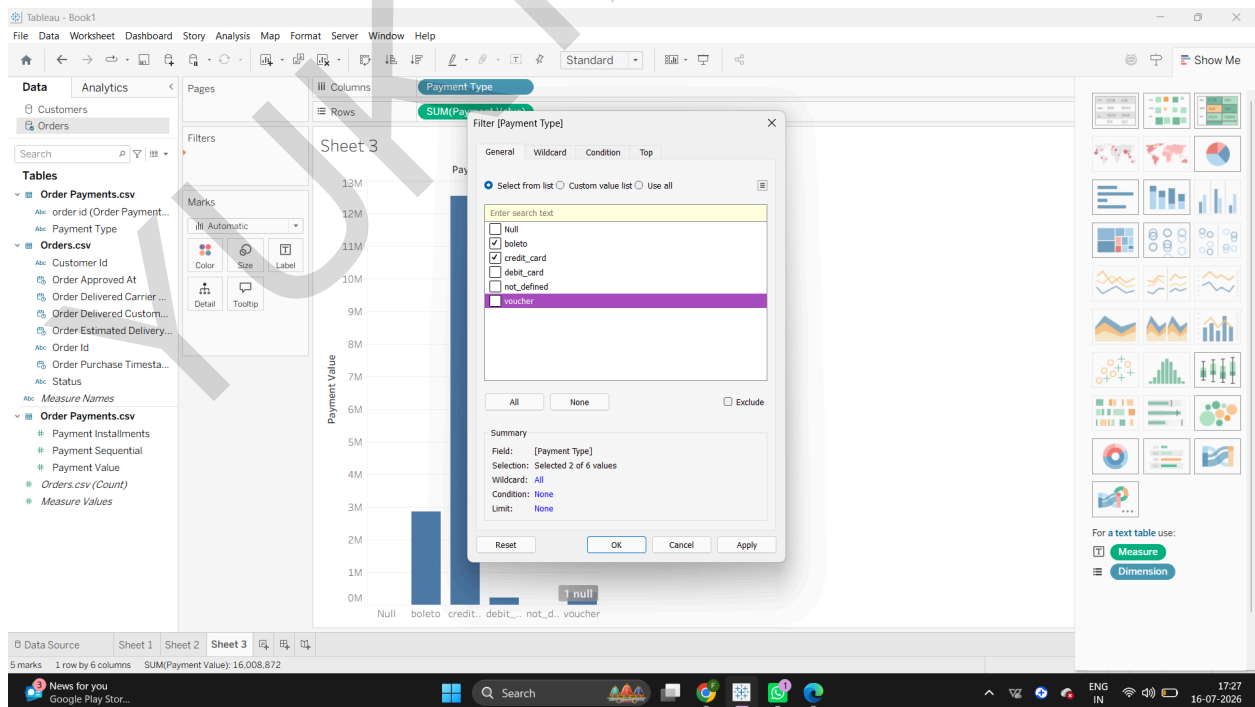
Step F: Implementing Multi-Tiered Filters

1. Open a clean worksheet. Drag the quantitative measure field payment_value onto the Rows shelf and place the categorical dimension payment_type onto the Columns shelf.

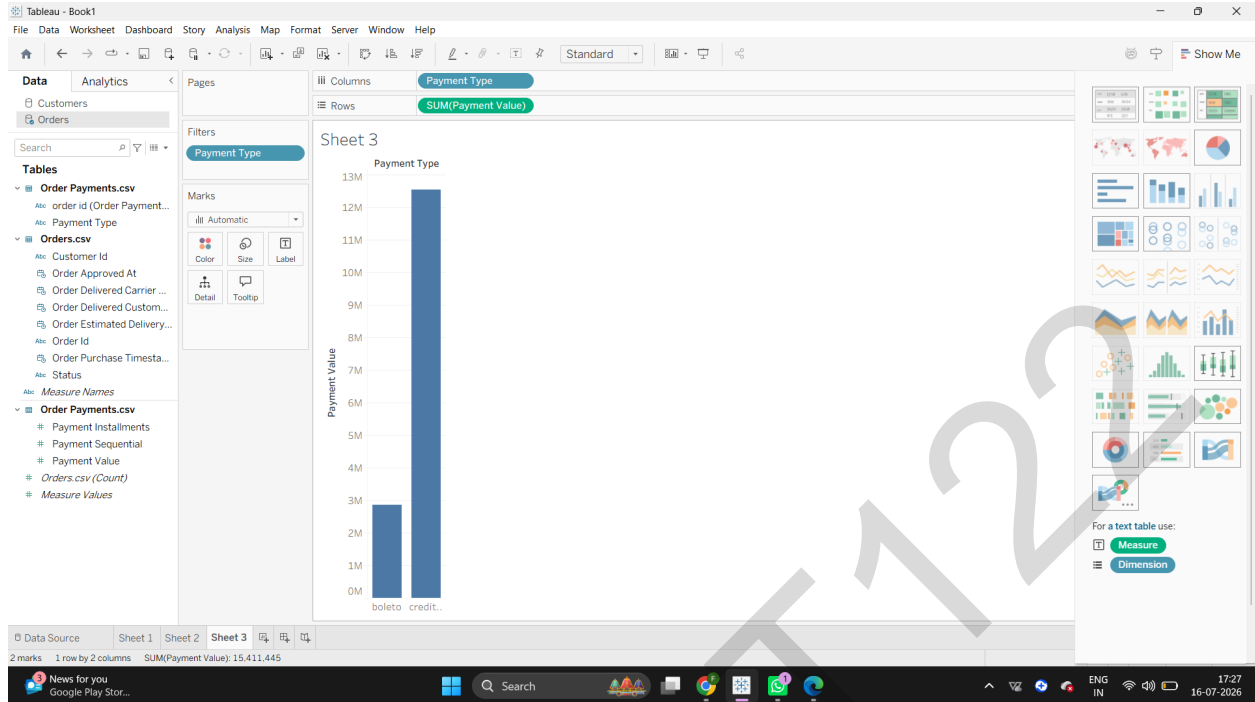
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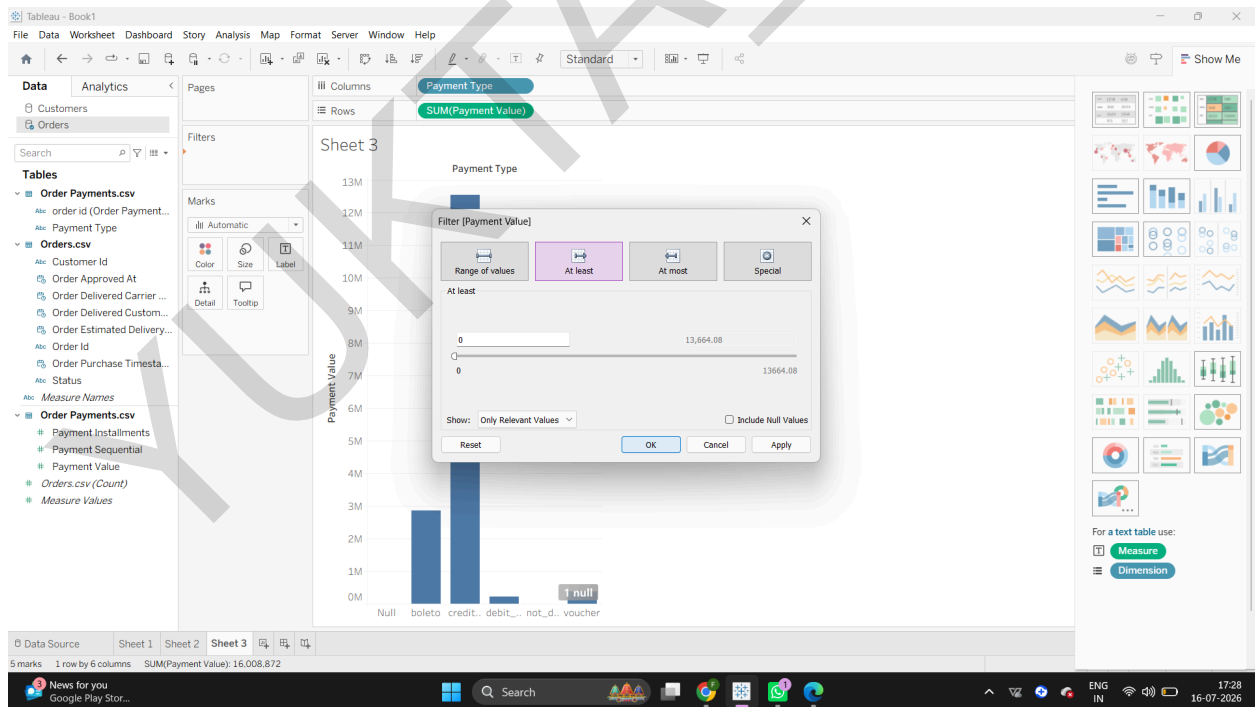
2. Dimension Filter: From your Data Pane list, drag the categorical string attribute payment_type and drop it onto the empty Filters shelf positioned immediately above the Marks card layout. Uncheck the box for not_defined, check the boxes for credit_card and boleto, and click OK.



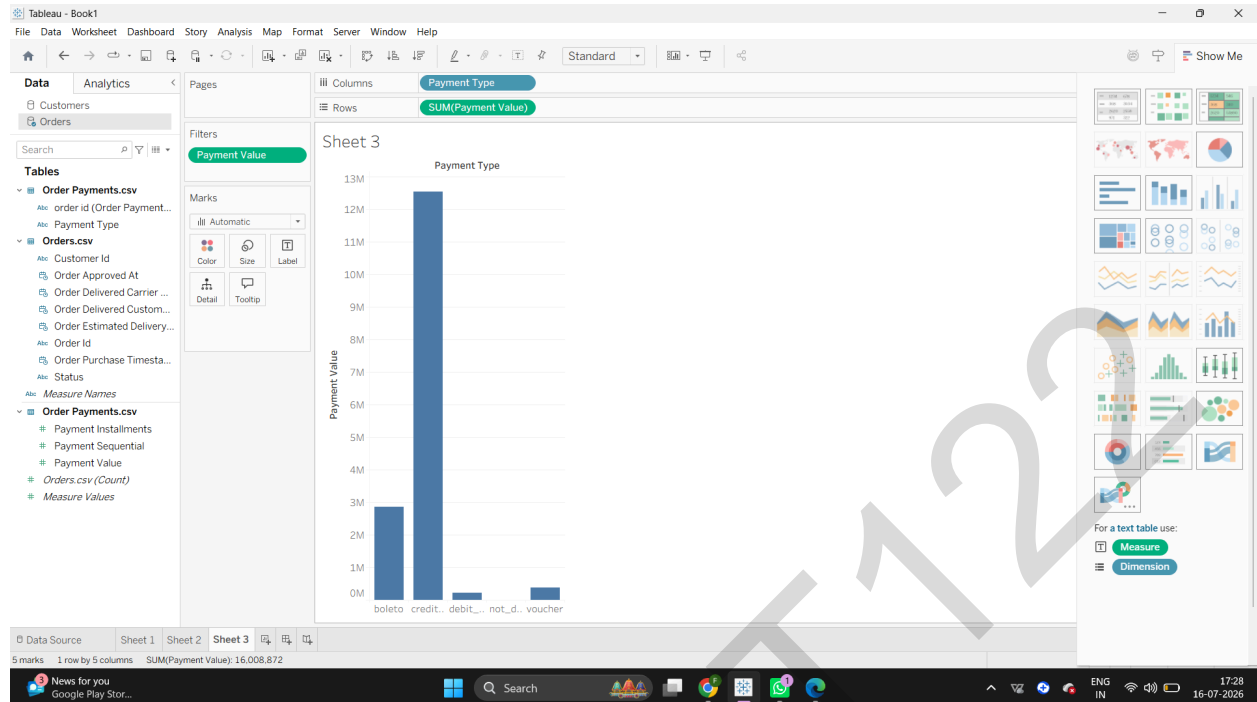
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3. Measure Filter: Drag the quantitative numeric variable payment_value out from the measures list and drop it directly onto the Filters shelf. Select All Values in the prompt window, click Next, select the At Least option condition, type in 50 to screen out micro transactions(it will be at 0.1 at by default Change it by clicking on it and typing 50 on it), and click OK.

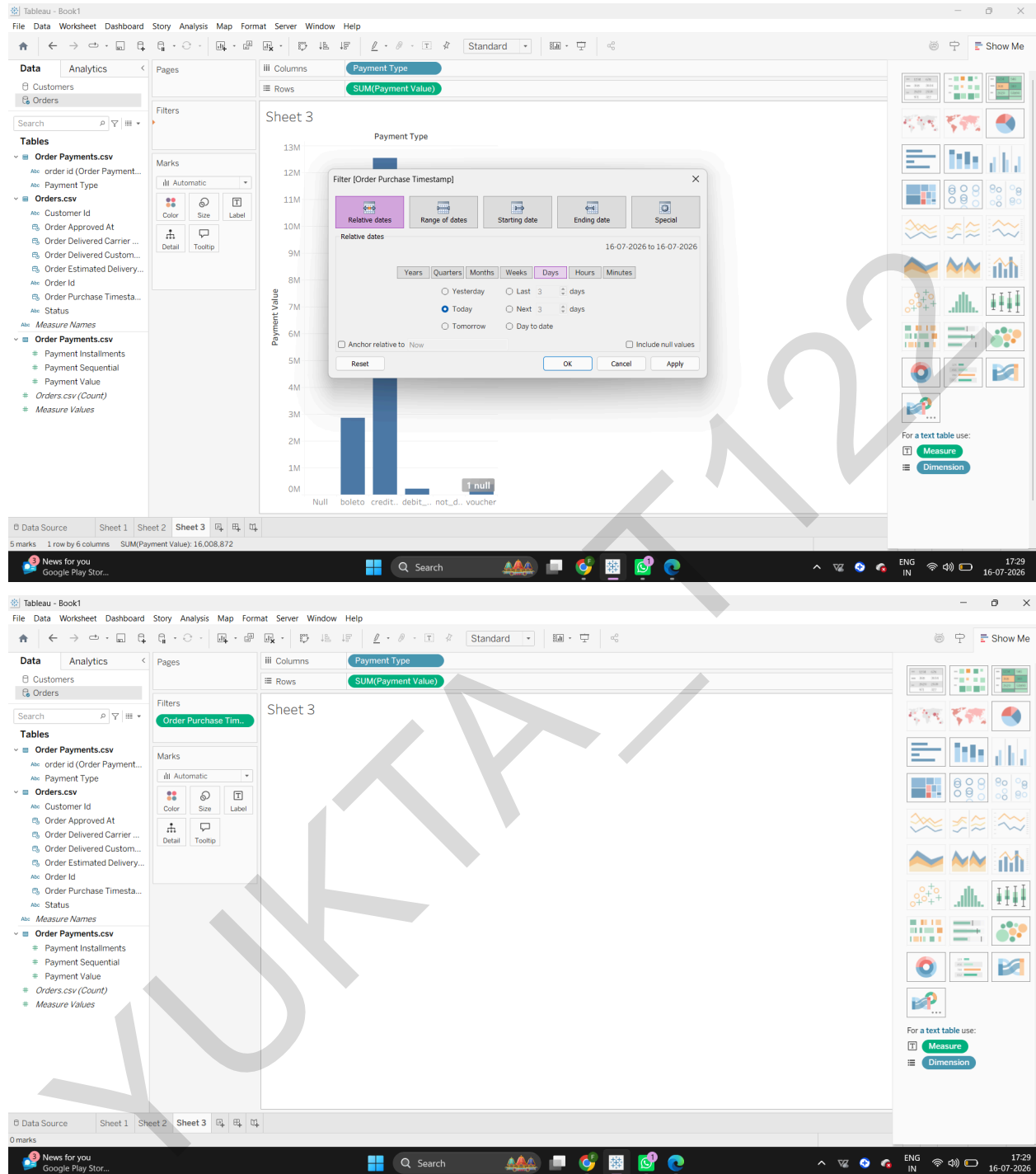


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4. Date Filter: Drag the calendar configuration time variable `order_purchase_timestamp` and drop it onto the Filters shelf. Select Years or Relative Date from the prompt configuration layout page to isolate the visualization into specific historical reporting boundaries.

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Step G: Optimizing Dashboard Performance 1. Extract Filters: Navigate back to the main Data Source configuration workspace screen. Look immediately below the selected Extract radio dial selection button in the upper-right area and click the small Edit link. Click Add, select the categorical field Status, choose

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to check delivered exclusively, and click OK. Tableau will purge all canceled or unfulfilled order rows from the local extract file entirely.

Tableau - Book1

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a9810da82917af2d9aefd127...	f2c7fc58a9de810828715166...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00
25e8ea4e93396b6fa0d3dd7...	25b14b69de0b6e184ae6fe27...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00
ba78997921bbcd1373bb41e...	7a5d8faaa1081f800628c30...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00
42fd880ba16b47b59251dd4...	15fd6fb8f8312dbb4674e451...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00
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0 Data Source Sheet 1 Sheet 2 Sheet 3

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Search

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Tableau - Book1

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Abc	Customer Id	Order...	custo...
Abc	Status	Order...	order_...

Customer Id	Status	Order Purchase Timestamp	Order Approved At
0a8556ac6be836b46b3e89...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00
f2c7fc58a9de810828715166...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00
25b14b69de0b6e184ae6fe27...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00
7a5d8faaa1081f800628c30...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00
15fd6fb8f8312dbb4674e451...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00
a24e6f72471e9dbafcb292bc...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00

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New Union
New Table Extension

Orders.csv 13 fields

Name Orders.csv

Description No description available.

Fields

Type	Field Name
Abc	Order Id
Abc	Customer Id
Abc	Status

Filter [Status]

General Wildcard Condition Top

Select from list Custom value list Use all

Enter search text

- ☐ approved
- ☐ canceled
- ☐ created
- ☒ delivered
- ☐ invoiced
- ☐ processing
- ☐ shipped
- ☐ unavailable

All None Exclude

Summary

Field: [Status]
Selection: Selected 1 of 8 values
Wildcard: All
Condition: None
Limit: None

Reset OK Cancel

100 rows

Customer Id	Status	Order Purchase Timestamp	Order Approved At
8556ac6be836b46b3e89...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00
7fc58a9de810828715166...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00
b14b69de0b6e184ae6fe27...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00
a5d8faaa1081f800628c30...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00
421df880ba16b47b59251dd4...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00
298fcd1f73eb413e4d26d01...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00

Data Source Sheet 1 Sheet 2 Sheet 3

30°C Mostly cloudy

Search

ENG IN

17:32 16-07-2026

Tableau - Book1

File Data Server Window Help

Connections

Orders

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Categories.csv
Customers.csv
Geolocation.csv
Order Items.csv
Order Payments.csv
Orders.csv
Products.csv
Reviews.csv
Sellers.csv

New Union
New Table Extension

Orders.csv 13 fields

Name Orders.csv

Description No description available.

Fields

Type	Field Name	Phys...	Rem...
Abc	Order Id	Order...	order_id
Abc	Customer Id	Order...	custo...
Abc	Status	Order...	order_...

Edit Data Source Filters

Filter Scope Details

Status Orders.csv and related t... keeps delivered

Add... Edit... Remove

OK Cancel

100 rows

Order Id	Customer Id	Status	Order Purchase Timestamp	Order Approved At
b81e226f3fe1789b1e8b2aca...	0a8556ac6be836b46b3e89...	delivered	25-04-2018 22:01:00	25-04-2018 22:15:00
a9810da82917af2d9aefd127...	f2c7fc58a9de810828715166...	delivered	26-06-2018 11:01:00	26-06-2018 11:18:00
25e8ea4e93396b6fa0d3dd7...	25b14b69de0b6e184ae6fe27...	delivered	12-12-2017 11:19:00	14-12-2017 09:52:00
ba78997921bbcd1373bb41e...	7a5d8faaa1081f800628c30...	delivered	06-12-2017 12:04:00	06-12-2017 12:13:00
421df880ba16b47b59251dd4...	15fd6fb8f8312dbb4674e451...	delivered	21-05-2018 13:59:00	21-05-2018 16:14:00
298fcd1f73eb413e4d26d01...	a24e6f72471e9dbafcb292bc...	delivered	07-05-2018 13:20:00	07-05-2018 15:31:00

Data Source Sheet 1 Sheet 2 Sheet 3

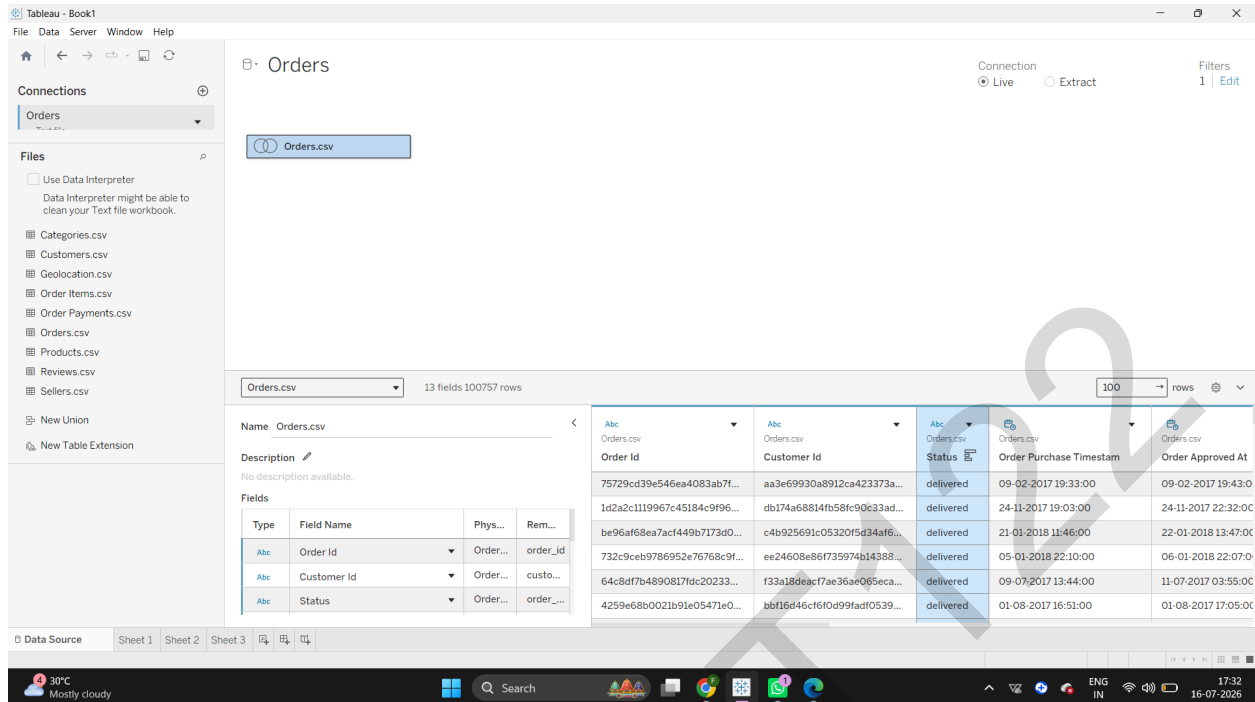
30°C Mostly cloudy

Search

ENG IN

17:32 16-07-2026

SHETH L.U.J. AND SIR M.V. COLLEGE



2. Context Filters: Return to an active worksheet layout canvas view. Move your cursor to your Filters shelf, right-click any active filter pill sitting in the shelf area, and select Add to Context from the context dropdown. The selected pill background formatting will immediately change to a dark gray color.

